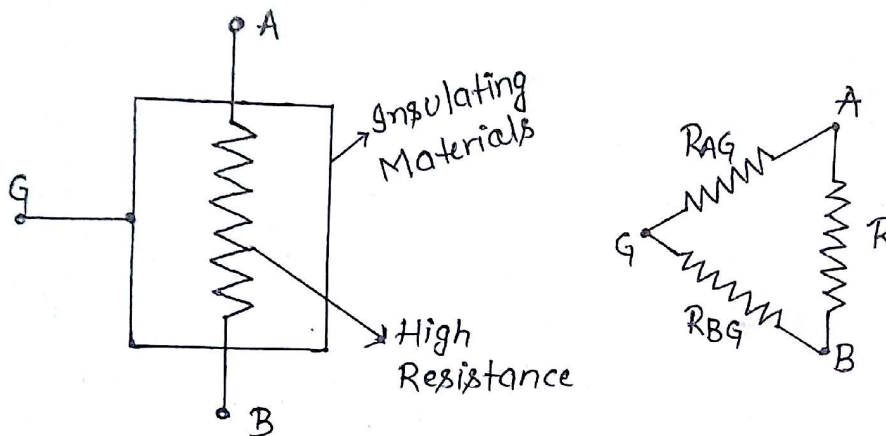


MEASUREMENT OF HIGH RESISTANCE

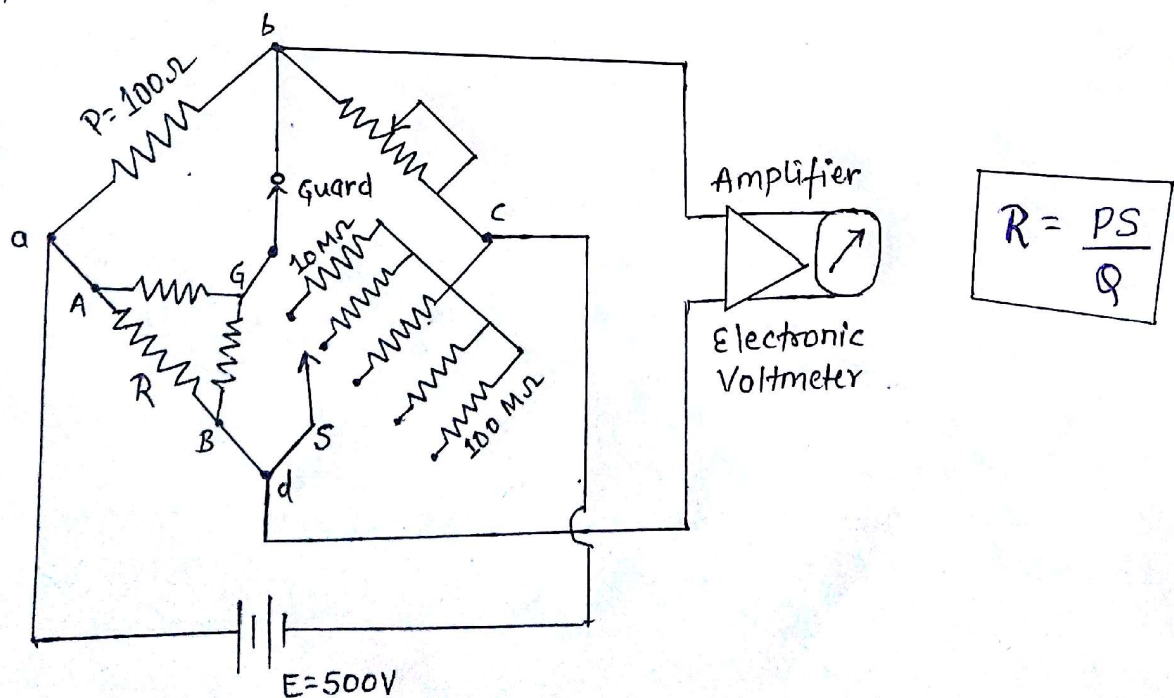
① Megohm Bridge :-

- Megohm bridge is an important method for measurement of high resistances. It has one three terminal high resistance located in one arm of the bridge.

Below fig show very high resistance with terminals A & B, and guard terminal which is put on the insulation. So it forms an equivalent three terminal resistance as shown below.



- If this high resistance is measured by ordinary wheatstone bridge it gives significant percentage error. So, we modify the wheatstone bridge for measurement of high resistances and this modified bridge is Megohm bridge, which gives less error.

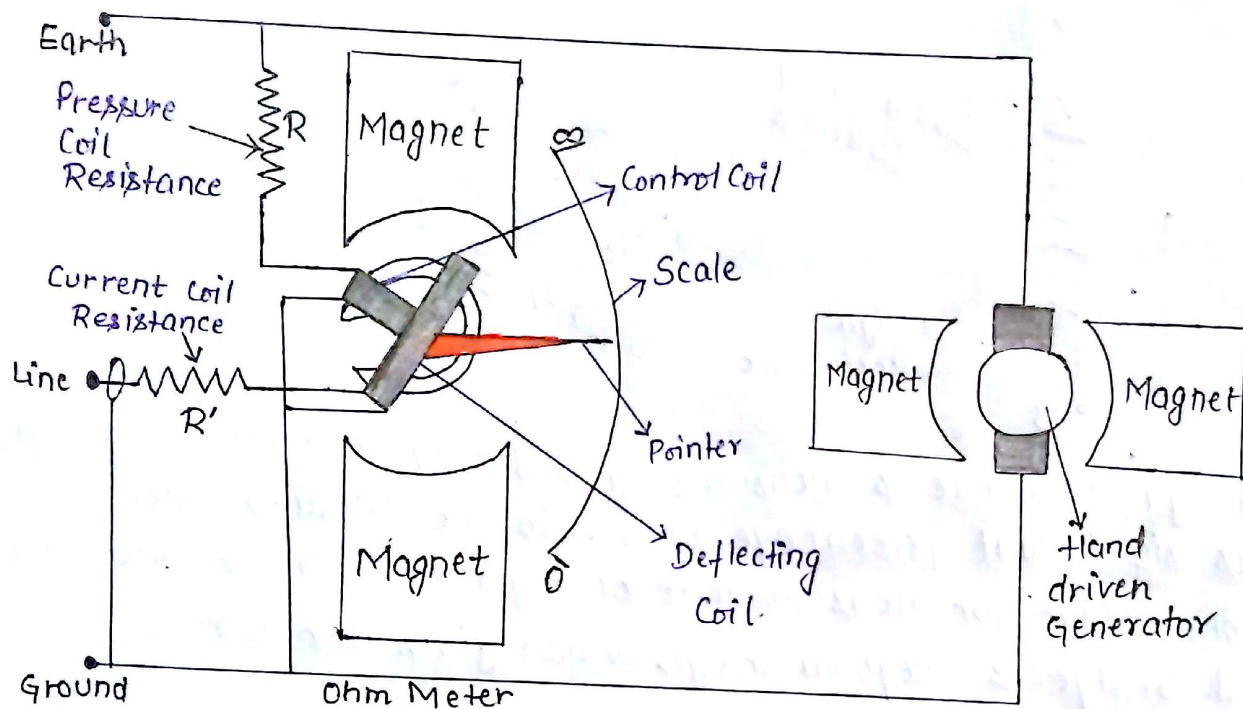


$$R = \frac{PS}{Q}$$

Megohm Bridge

- Sensitivity for balancing against high resistance is obtained by use of adjustable high voltage supplies of 500 V or 1000 V and use of sensitive null indicating arrangement such as high gain amplifier with electronic Voltmeter or a CRO.
- Megohm has a range from $0.1 \text{ M}\Omega$ to $10^6 \text{ M}\Omega$ and the accuracy is within 3% for lower part and 10% above 10000 $\text{M}\Omega$.

② Megger :-



* Deflecting and Control Coil :- Both are connected in parallel to the generator, mounted at right angle to each other and maintain polarities in such a way that produced torque in opposite direction.

* Permanent Magnets :- Produce magnetic field to deflect pointer with North-South pole magnets.

* Pointer :- One end of pointer is connected to the coil and another end deflects on scale from infinity to zero.

* DC generator :- Testing voltage is produced by hand operated DC generator for manual operated Meggar.

* Pressure Coil Resistance & Current Coil Resistance :-

Protect instrument from any damage because of low electrical external resistance under test.

WORKING OF MEGGER :-

- * Working is based on ohm-meter or ratio-meter. The deflection torque is produced with megger tester due to magnetic field produced by voltage and current, similar like 'Ohm's law'.
 - * Torque of megger varies in ratio with V/I . Electrical resistance to be measured is connected across generator and in series with deflecting torque coil.
 - * High Res~~is~~ Produced torque shall be in opposite direction if current supplied to the coil.
1. High Resistance (No current) :- No current shall flow through deflecting coil, if resistance is very high i.e. infinity position of pointer.
 2. Small Resistance (High current) :- If circuit allow a high current to pass through deflecting coil for measuring small resistance i.e. produced torque makes pointer to set at 'ZERO'.
 3. Intermediate Resistance (Varied current) :- If measured resistance is intermediate, produced torque align or set the pointer between the range of 'ZERO' to 'INFINITY'.

A.C. BRIDGES

AC bridges are those circuits which are used to measure the unknown resistances, capacitance, inductance, frequency and mutual inductance.

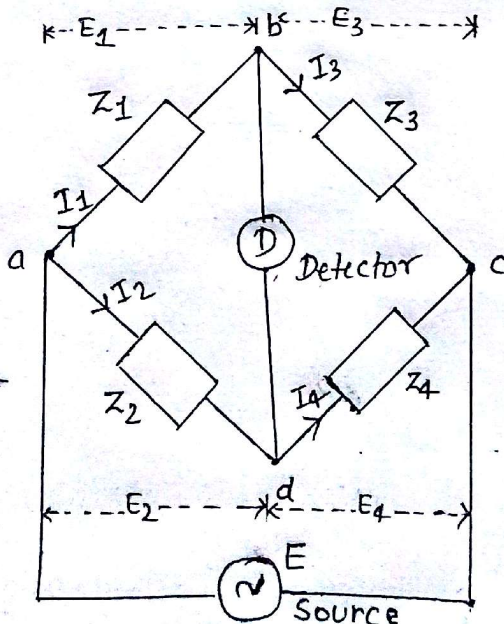
When bridge is balanced, detector can be removed.

and

$$\frac{E_1}{E_2} \neq \frac{E_3}{E_4}$$

$$\frac{Z_1}{Z_2} = \frac{Z_3}{Z_4}$$

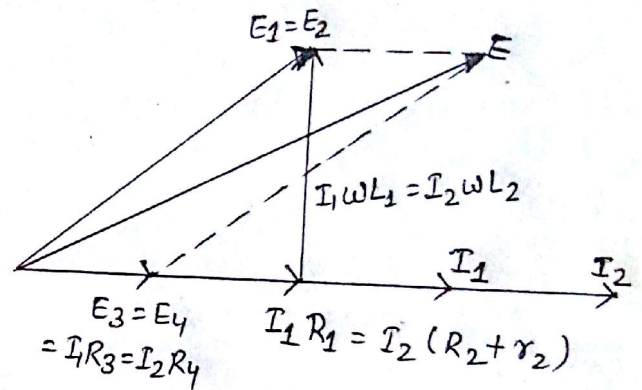
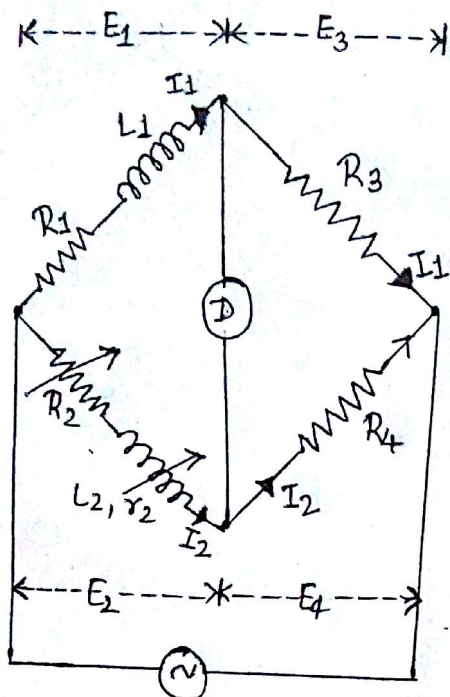
Basic
Bridge
Network



MEASUREMENT OF INDUCTANCE :-

① Maxwell's Inductance Bridge :-

- * In the Maxwell's inductance bridge, there are two pure resistances used for balance relations but on other side or arms two impedances are used.
- * This bridge circuit measures an inductance by comparison with a variable standard self-inductance.



When balanced, we have $\frac{Z_1}{Z_2} = \frac{R_3}{R_4} \Rightarrow \frac{R_1 + j\omega L_1}{(R_2 + r_2) + j\omega L_2} = \frac{R_3}{R_4}$

$$(R_1 + j\omega L_1)R_4 = [(R_2 + r_2) + j\omega L_2]R_3$$

Separating Real and imaginary parts, we get

$$\boxed{L_1 = L_2 \frac{R_3}{R_4}} \quad \text{and} \quad \boxed{R_1 = \frac{R_3}{R_4} (R_2 + r_2)}$$

② Maxwell's Inductance Capacitance Bridge :-

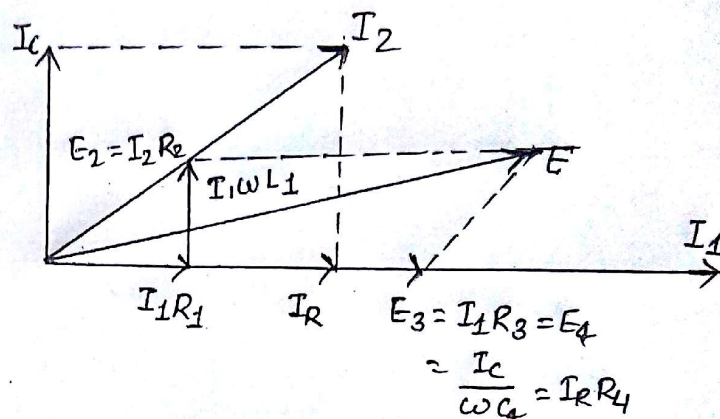
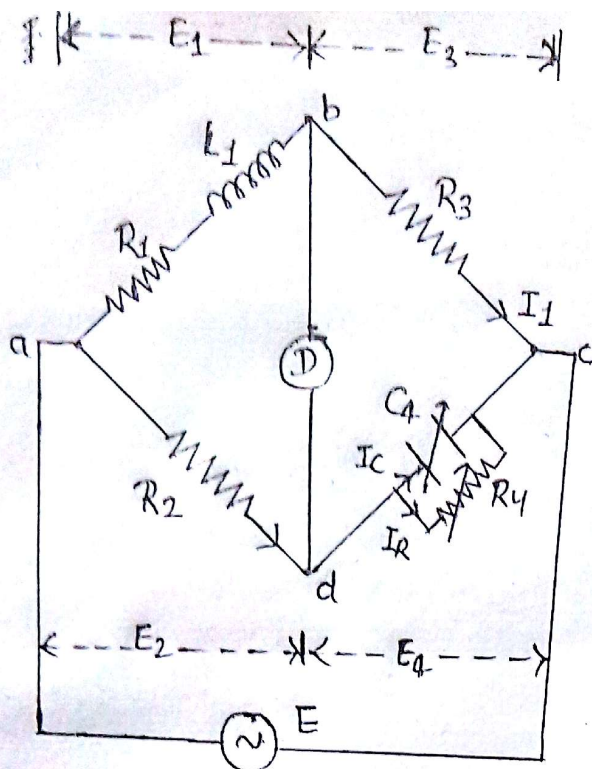
In this bridge, an inductance is measured by comparison with a standard variable capacitance.

Let, L_1 = unknown inductance

R_1 = effective resistance of inductor L_1 ,

R_2, R_3, R_4 = known resistance

C_4 = variable standard capacitor,



Maxwell's Inductance capacitance bridge and its phasor diagram

On balanced condition , $Z_1 Z_4 = Z_2 Z_3$

$$\Rightarrow (R_1 + j\omega L_1) \left(\frac{R_4}{1 + j\omega C_4 R_4} \right) = R_2 R_3 \quad \text{or} \quad R_1 R_4 + j\omega L_1 R_4 = R_2 R_3 + j\omega R_2 R_3 C_4 R_4$$

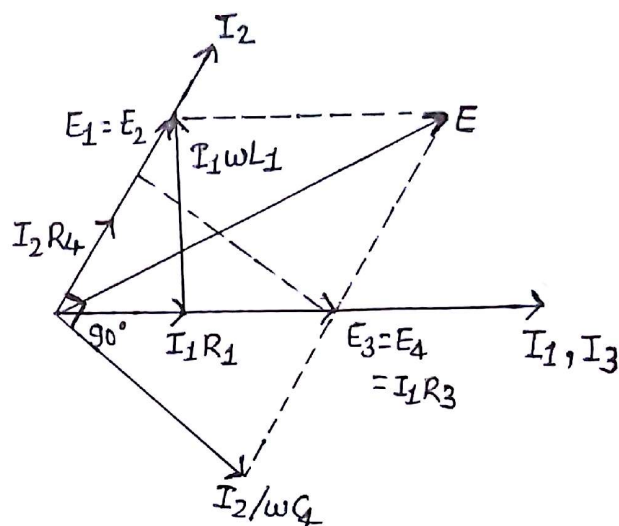
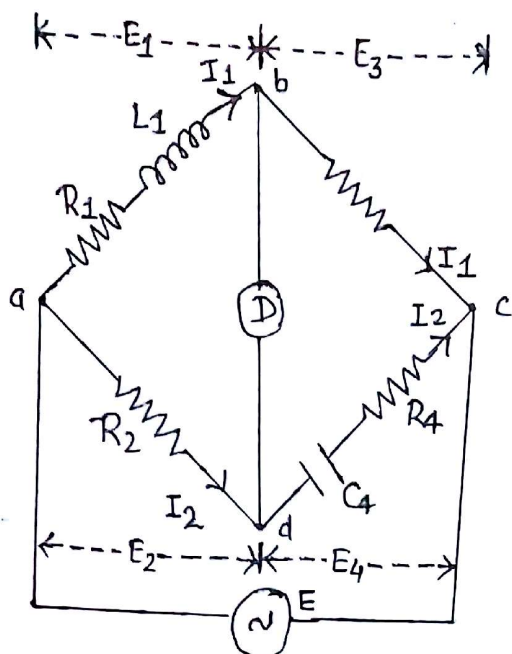
Separating Real and imaginary part, we get

$$\boxed{R_1 = \frac{R_2 R_3}{R_4}} \quad \& \quad \boxed{L_1 = R_2 R_3 C_4}$$

- * This bridge requires a variable standard capacitor which may be very expensive if calibrated to high degree of accuracy.
- * The bridge is limited to measurement of low quality factor coils.
($1 < Q < 10$) $\boxed{Q = \omega L_1 / R_1 = \omega C_4 R_4}$

③ Hay's Bridge :-

- The Hay's Bridge is modification of Maxwell's bridge. This bridge uses a resistance in series with standard capacitor.
- This bridge is particularly useful if phase angle of inductive impedance is large.



Hay's Bridge and its Phasor Diagram

At balanced condition, $Z_1 Z_4 = Z_2 Z_3$
 $\Rightarrow (R_1 + j\omega L_1)(R_4 - j/\omega C_4) = R_2 R_3$ or $R_1 R_4 + \frac{L_1}{C_4} + j\omega L_1 R_4 - \frac{j R_1}{\omega C_4} = R_2 R_3$

Separating real and imaginary part, we get

$$R_1 R_4 + \frac{L_1}{C_4} = R_2 R_3 \quad \& \quad L_1 = -\frac{R_1}{\omega^2 C_4 R_4}$$

By solving this, we get $L_1 = \frac{R_2 R_3 C_4}{1 + \omega^2 C_4^2 R_4^2}$ & $R_1 = \frac{\omega^2 R_2 R_3 R_4 C_4^2}{1 + \omega^2 C_4^2 R_4^2}$

Also, Q factor $\Rightarrow Q = \frac{\omega L_1}{R_1} = \frac{1}{\omega C_4 R_4}$

* Inductance and resistance contains the frequency term. Hence it appears that the frequency of source of supply to the bridge must be accurately known.

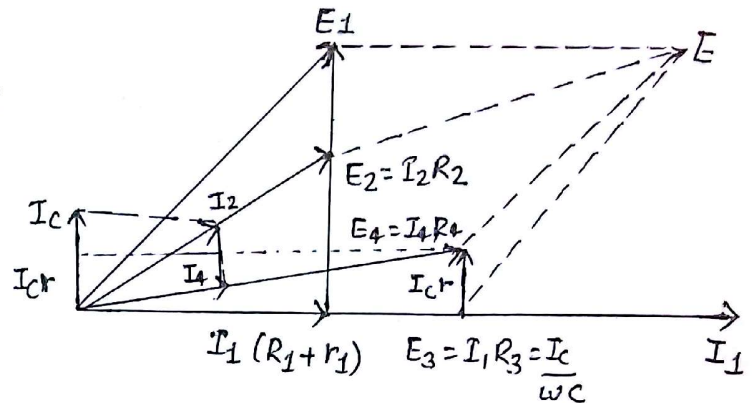
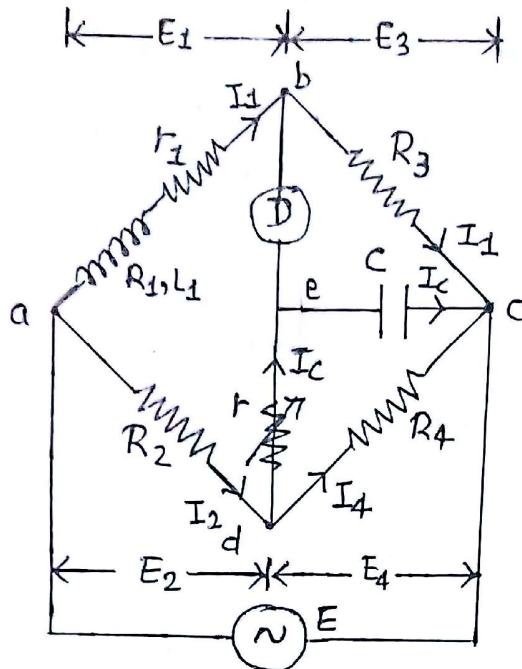
* for high Q-factor inductance ~~and resistance~~ becomes frequency independent as, $L_1 = \frac{R_2 R_3 C_4}{1 + \omega^2 C_4^2 R_4^2} = \frac{R_2 R_3 C_4}{1 + (1/Q)^2}$

if $1 \gg (1/Q)^2$ $L_1 = R_2 R_3 C_4$

Now, this inductance is same as that of inductance measured by Maxwell's Inductance Capacitance bridge.

④ Anderson's Bridge :-

- * In the Anderson Bridge the unknown inductance is measured in terms of known capacitance and resistance.
- * This method is applicable for precise measurement of self-inductance over a very wide range of values.



Anderson's Bridge and its Phasor Diagram

At balanced conditions, $I_1 = I_3$ and $I_2 = I_c + I_4$

$$\text{Now, } I_1 R_3 = I_c \times \frac{1}{j\omega C} \Rightarrow \boxed{I_c = I_1 j\omega C R_3} \rightarrow \textcircled{1}$$

Other balanced eqns,

$$I_1 (r_1 + R_1 + j\omega L_1) = I_2 R_2 + I_c r \quad \& \quad I_c \left(r + \frac{1}{j\omega C} \right) = (I_2 - I_c) R_4$$

from $\textcircled{1}$, we have

$$I_1 (r_1 + R_1 + j\omega L_1) = I_2 R_2 + I_1 j\omega C R_3 r \quad \text{or} \quad I_1 (r + R_1 + j\omega L_1 - j\omega C R_3 r) = I_2 R_2 \quad \textcircled{2}$$

$$\& \quad j\omega C R_3 I_1 \left(r + \frac{1}{j\omega C} \right) = (I_2 - j\omega C R_3 I_1) R_4 \quad \text{or} \quad I_1 (j\omega C R_3 r + j\omega C R_3 R_4 + R_3) = I_2 R_4 \quad \textcircled{3}$$

from $\textcircled{2}$ & $\textcircled{3}$, we get

$$I_1 (r_1 + R_1 + j\omega L_1 - j\omega C R_3 r) = I_1 \left(\frac{R_2 R_3}{R_4} + j\omega C \frac{R_2 R_3 r}{R_4} + j\omega C R_3 R_2 \right)$$

separating real & imaginary part, we get

$$\boxed{R_1 = \frac{R_3 R_2}{R_4} - r_1}$$

&

$$\boxed{L_1 = C \frac{R_3}{R_4} [r(R_2 + R_4) + R_2 R_4]}$$